

Quantum Dice User Manual

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Introduction

Once your Quantum Dice assembly is complete, you're ready to start using it! This manual covers:

- Powering the device on and off
- Understanding the startup sequence
- Using the Quantum button
- Entangling two quantum dice
- Charging the battery

Device orientation: The dice has 3 axes, color-coded as follows:

- **Blue** - top and bottom sides
 - **Red** - left and right sides
 - **Yellow** - front and rear sides Watch the [instructional video](#) for usage guidance
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Power On

Locate the power button on one of the yellow sides (marked with a power symbol).

To power on/off: Press the button briefly.

Charging the Battery

Battery specifications:

- Capacity: 1800 mAh
- Charging current: 500 mA

Charging instructions:

1. Locate the USB-C connector on one of the yellow sides

2.  **Important:** This connector is **only** for battery charging and is not connected to the ESP32 processor
3. **Use a USB-A to USB-C cable** with a USB-A charger (USB-C chargers will not work)
4. Charging takes approximately **2-3 hours**

Monitoring charging status:

The charge indicator LEDs are located inside the dice and are not visible during charging. To monitor the charging process, you can use:

- A USB charger with a built-in current indicator, or
 - A USB current monitor dongle such as [this one](#)
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Start Up Sequence

After powering on, the dice initializes and displays the following:

1. **Qlab logo** followed by additional branding
2. **Device information** on the power button display:
 - Dice ID
 - Firmware version
 - Battery voltage

 **Low battery warning:** If the battery voltage is low, charge the device before use. See [Charging the Battery](#)

The startup sequence completes in **Classic mode**, where each side displays a fixed number (like a traditional die). Press the Quantum button to enter Quantum mode.

Basic Quantum State

In Quantum mode, all six sides display a **superposition state**.

How it works:

- Roll the dice
 - When it comes to rest, a random number appears on the top and bottom sides
 - This represents the "measurement" of the quantum state
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Using the Quantum Button

The Quantum button is located on the opposite side from the power button.

Long press: Clears the previous measurement and returns the dice to the superposition state.

Entanglement

Requirements: Both dice must be in Quantum mode and have **matching IDs**.

How to entangle:

1. Ensure both dice are in Quantum mode (superposition state visible)
2. Bring the two dice close together
3. You may need to adjust their position slightly
4. **Success indicator:** The superposition state turns **yellow**

After entanglement:

- Roll both dice separately
- The measurement outcomes will be **correlated** between the two dice
- After measurement, each die returns to its previous state

 **Important:** After entanglement, separate the dice by **more than 1 meter** to prevent them from re-entangling after rolling.

Enjoy your Quantum Dice!